

1. Introduction to free electron theory:

In the realm of solid-state physics, the behaviour of electrons in materials is a topic of great importance. Understanding how electrons move and interact in a solid is essential for explaining various electrical, thermal, and optical properties of materials. The free electron theory provides a simplified yet insightful model for comprehending the behaviour of electrons in metals. The electron theory has been developed in three main stages.

- (i) Classical free electron theory.
- (ii) Quantum free electron theory.
- (iii) Zone theory.

(i) **Classical free electron theory or (Drude and Lorentz theory of free electrons in metals)**

The Classical Free Electron Theory is a simplified model used in solid-state physics to describe the behaviour of electrons in metals. **Drude and Lorentz** introduced the initial theory in 1900. According to this theory, metals contain free electrons that account for electrical conductivity, and these electrons adhere to the principles of classical mechanics.

(ii) **Assumptions of Classical free electron theory (Drude and Lorentz theory of free electrons in metals):**

- (a) **Free Electron Assumption:** Electrons in a metal are assumed to be essentially free and not bound to specific atoms. They are considered to move freely throughout the entire metal lattice, experiencing the collective forces of the positively charged atomic cores.
- (b) **Neglect the Electron-Electron Interactions:** In this theory, electrons are treated as non-interacting particles, which simplifies the mathematical description of their behaviour. While this simplification is useful for some purposes, it does not account for electron-electron scattering events that can occur.
- (c) **Electron Gas Model:** This theory assumes the free electrons within the metal behave like gas molecules in the container and obeying the laws of kinetic theory of gases.
- (d) **Continuous Energy Levels:** Unlike electrons in isolated atoms, which have discrete energy levels, in the Classical Free Electron Theory, the energy levels of free electrons are treated as a continuum.
- (e) **Drift Velocity and Ohm's Law:** When an electric field is applied to a metal, the free electrons are assumed to respond to this field by drifting in the direction of the field. This drift of electrons constitutes an electric current, and the relationship between the applied electric field, drift velocity, and current density is described by Ohm's law.
- (f) Electron velocities in a metal obey Maxwell-Boltzmann distribution of velocities.

(g) In the absence of any external field the energy associated with each electron at temperature T is given by $E = \frac{3}{2} k_B T$, where k_B is Boltzmann constant.

(h) The electron move in a constant potential field.

(iii) **Advantages and disadvantages of classical free electron theory:**

Advantages:

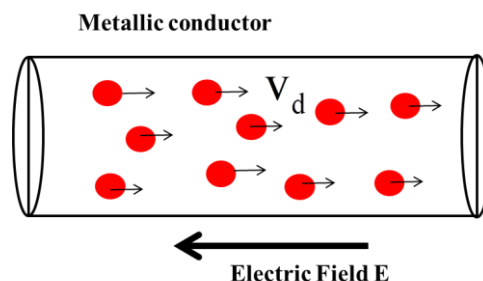
- i. It predicts the electrical conductivity and verifies the Ohm's law.
- ii. It gives the intuitive framework to understand the basic electrical and thermal properties of metals.
- iii. It explains optical properties of metals.
- iv. Derivation of the Widemann-Franz law.

Disadvantages:

- i. It fails to describe the behaviour of semiconductors, insulators, and other complex materials.
- ii. It fails to explain the changes in electrical conductivity with temperature, especially at low temperatures.
- iii. It does not provide an explanation for the specific heat of metals.
- iv. Classical free electron theory cannot account for phenomena such as the photoelectric effect, the Compton Effect, and black body radiation.
- v. It lacks to give an explanation for the temperature-dependent behaviours of paramagnetic susceptibility and ferromagnetism.

Definitions of quantities used in classical free electron theory:

1. **Drift velocity (v_d):** It is the average velocity attained by an electron under the influence by an electric field.



Suppose an electron of mass m , is set to move in an electric field E . Due to this electric field the electron will experience an electric force $F = eE$, opposite to the direction of electric field. Due to this force electron attains the velocity v_d

Now from Newton's second Law of motion:

$$F = ma = m \frac{d^2x}{dt^2} \dots \dots \dots (1)$$

Since we know that

$$F = eE \dots \dots \dots (2)$$

From equations (1) and (2), we can write

$$m \frac{d^2x}{dt^2} = eE$$

$$\Rightarrow \frac{d^2x}{dt^2} = \frac{eE}{m}$$

Integrating above equation, we have

$$\Rightarrow \frac{dx}{dt} = \frac{eE}{m} t + C$$

Since at $t = 0 \Rightarrow \frac{dx}{dt} = 0$

$$\Rightarrow C = 0$$

Therefore we have

$$v_d = \frac{dx}{dt} = \frac{eE}{m} \tau \dots \dots \dots (3)$$

Where $t = \tau$, is the relaxation time of the electron.

2. **Relaxation Time(τ):** Under the influence of external electric field, electrons attain the directional velocity v_d and if the electric field is switched off, velocity starts decreasing exponentially. Such a process which tends to restore the equilibrium is called **relaxation process**.

If $v_0 = v_d$ is the velocity at $t = 0$ at which the field is switch off. Now the velocity of the electron will starts decreasing exponentially.

$$v = v_d e^{-\frac{t}{\tau}}$$

Therefore the time taken by an electron to reduce its velocity $\frac{1}{e} v_d$ after the switch off of electric field is called relaxation Time.

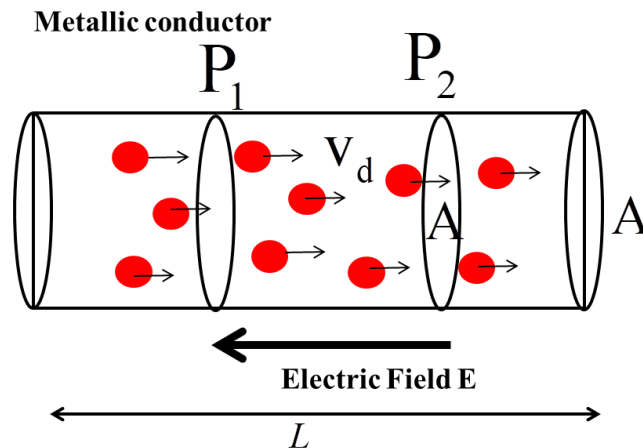
3. **Mean Free Path(λ):** The average distance travelled by the electrons between two successive collisions is known as mean free path.

4. **Mean Collision Time(τ_c):** Average time taken by the electron between two successive collisions.

$$\tau_c = \frac{\lambda}{v}$$

Where λ , is the mean free path and v is the velocity of the electron.

5. **Electrical conductivity in metals(σ):**



Let us consider a conductor of length L and cross-sectional area A .

So the Volume of the conductor = AL

Suppose the number of free electrons per unit volume = n

Therefore the total number of electrons in the conductor = nAL

Thus the total charge in the conductor $Q = enAL$

Suppose the time taken by the electrons to move from one end to another.

Therefore the current i in the conductor,

$$i = \frac{Q}{t} = \frac{enAL}{t} \dots \dots \dots (1)$$

Since the electrons moves in the conductor with drift velocity v_d ,

$$\text{Therefore } v_d = \frac{L}{t} \dots \dots \dots (2)$$

Thus the equation (1) can be written as

$$i = n e A v_d \dots \dots \dots (3)$$

Current Density (J): It is defined as the current per unit of area.

$$J = \frac{i}{A} = n e v_d \dots \dots \dots (4)$$

Again from the microscopic form of Ohm's Law

$$J = \sigma E \dots \dots \dots (5)$$

Where, σ , is the electrical conductivity of the conductor, and E , is applied electric field.

From equations (4) and (5), we can write

$$\sigma = \frac{J}{E} = \frac{n e v_d}{E} \dots \dots \dots (6)$$

Since we know that $v_d = \frac{dx}{dt} = \frac{eE}{m} \tau \dots \dots \dots (7)$

Therefore we can write

$$\sigma = \frac{J}{E} = \frac{n e v_d}{E} = \frac{n e^2 \tau}{m} \dots \dots \dots (8)$$

Unit of electrical conductivity is ***ohm⁻¹ – meter⁻¹ or siemens/meter***

Resistivity(ρ): Resistivity of a conductor is the reciprocal of the electrical conductivity.

$$\rho = \frac{1}{\sigma} = \frac{m}{n e^2 \tau} \dots \dots \dots (9)$$

Unit of resistivity is ***ohm-meter***

Mobility of the electrons: It is defined as drift velocity per unit electric field.

$$\mu = \frac{v_d}{E} = \frac{\sigma}{n e} \dots \dots \dots (10)$$

$$\Rightarrow \sigma = n e \mu \dots \dots \dots (11)$$

Unit of mobility ***meter²/Volt.Sec***

6. Temperature dependence of electrical conductivity:

Since we know that electrical conductivity of the metal is given by (as discussed in equation (8))

$$\sigma = \frac{n e^2 \tau}{m} \dots \dots \dots (12)$$

Further we know that Average energy of an electron is given by

$$E_{ave} = \frac{3}{2} k_B T \dots \dots \dots (13)$$

This average energy in the electron, will be in the form of kinetic energy,

$$E_{ave} = \text{K.E.} = \frac{mv_{th}^2}{2} \dots \dots \dots (14)$$

Where v_{th} , is the thermal velocity of the electron at temperature T, and it is given by

$$v_{th} = \sqrt{\frac{3k_B T}{m}} \dots \dots \dots (15)$$

Now the collision time is inversely proportional to thermal velocity

$$\tau \propto \frac{1}{v_{th}}$$

$$\Rightarrow \tau \propto \frac{1}{\sqrt{T}}$$

Therefore from equation (1) (Theoretical prediction)

$$\sigma \propto \frac{1}{\sqrt{T}} \dots \dots \dots (16)$$

But experimental observation says that

$$\sigma \propto \frac{1}{T} \dots \dots \dots (17)$$

Hence from equations (16) & (17), prediction of classical free electron theory is not in agreement with the experimental observations. Thus the classical free electron theory fails to explain dependence of T on σ .

(ii) **Quantum free electron theory:**

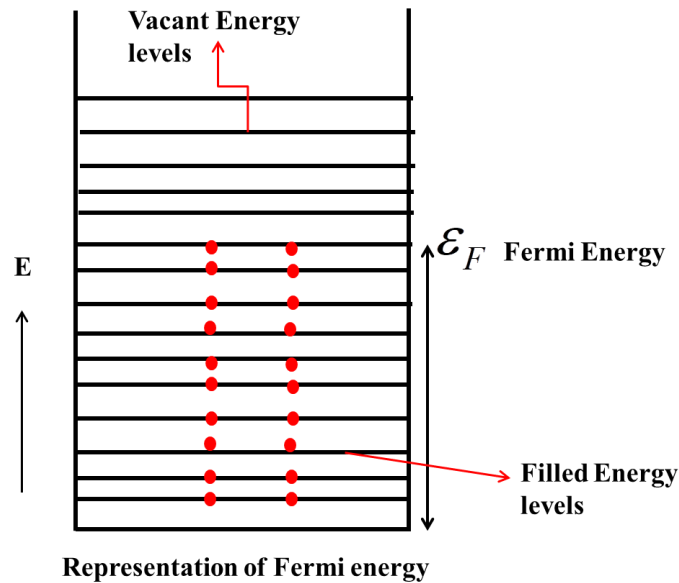
In 1928, **Sommerfeld** introduced the Quantum Free Electron Theory, wherein he posited that free electrons exhibit motion within a uniform potential. This theory adheres to the principles of quantum mechanics. The electrons in this model are treated as if they move in a potential energy well created by the positively charged atomic cores. In this potential well, electrons can have discrete energy levels, analogous to quantum states.

Assumptions of quantum free electron theory:

1. Free electrons in a metal possess discrete energy levels, which leads to energy quantization.
2. Electrons adhere to Pauli's Exclusion Principle, limiting the occupancy of any energy level to a maximum of two electrons.
3. The electron distribution across various energy levels complies with Fermi-Dirac quantum statistics.
4. Electrons within a metal generally experience consistent potential energy throughout because of the uniform potential created by ionic cores.
5. The forces of attraction between electrons and lattice ions, as well as the repulsion between electrons, are typically considered negligible.
6. Electrons are commonly described as exhibiting wave-like properties in addition to their particle-like nature.

Fermi Dirac Statistics:

1. Fermi-Dirac statistics incorporates the Pauli Exclusion Principle, which states that no two identical fermions (particles with half-integer spin) can occupy the same quantum state simultaneously. As a result, it is impossible for multiple electrons in a system to have the same set of quantum numbers.
2. Particle Distribution: Fermi-Dirac statistics describe the probability of finding particles in different energy levels. It shows that at low temperatures, particles tend to occupy the lowest available energy states, and as the temperature increases, more energy levels become populated.
3. **Fermi Energy:** *The Fermi energy ϵ_F is a critical parameter in Fermi-Dirac statistics. It represents the highest energy level occupied by electrons at absolute zero ($T = 0\text{ K}$) temperature.*



4. The distribution function defined by Fermi-Dirac statistics depends on temperature.

$$f(\epsilon) = \frac{1}{e^{\frac{(\epsilon - \epsilon_F)}{k_B T}} + 1} \dots \dots \dots (1)$$

The expression given by equation (1) is also called Fermi factor.

At absolute zero ($T = 0 \text{ K}$), all energy states below the Fermi energy are occupied, and those above are unoccupied. As temperature increases, more states above the Fermi energy become occupied.

Variation of Fermi Factor or Fermi distribution function with energy and temperature:

(i) Case I: for $\epsilon < \epsilon_F$ at $T = 0 \text{ K}$

Value of Fermi factor

$$f(\epsilon) = \frac{1}{e^{\frac{(\epsilon - \epsilon_F)}{k_B T}} + 1} \dots \dots \dots (1)$$

For $\epsilon < \epsilon_F$ and $T = 0 \text{ K}$

$$(\epsilon - \epsilon_F) < 0$$

Which means that numerator of $\frac{(\epsilon - \epsilon_F)}{k_B T}$ **will always be negative** and $\frac{1}{T} = \frac{1}{0} = \infty$.

Therefore

$$f(\epsilon) = \frac{1}{e^{\frac{(\epsilon - \epsilon_F)}{k_B T}} + 1} = \frac{1}{e^{-\infty} + 1} = 1$$

$$f(\epsilon) = 1$$

Which means that $f(\epsilon) = 1$ for $\epsilon < \epsilon_F$ at $T = 0 \text{ K}$

This implies that at absolute zero ($T = 0K$) temperature, all the energy levels below fermi level are 100% occupied which is true from the definition of Fermi energy.

(ii) Case II: for $\epsilon > \epsilon_F$ at $T = 0K$

Value of Fermi factor

$$f(\epsilon) = \frac{1}{e^{\frac{(\epsilon - \epsilon_F)}{k_B T}} + 1} \dots \dots \dots (1)$$

For $\epsilon > \epsilon_F$ and $T = 0K$

$$(\epsilon - \epsilon_F) > 0$$

Which means that numerator of $\frac{(\epsilon - \epsilon_F)}{k_B T}$ will always be positive and $\frac{1}{T} = \frac{1}{0} = \infty$.

Therefore

$$f(\epsilon) = \frac{1}{e^{\frac{(\epsilon - \epsilon_F)}{k_B T}} + 1} = \frac{1}{e^{\infty} + 1} = 0$$

$$f(\epsilon) = 0$$

Which means that $f(\epsilon) = 0$ for $\epsilon > \epsilon_F$ at $T = 0K$

This implies that at absolute zero ($T = 0K$) temperature, all the energy levels above the fermi level are unoccupied or completely empty which is true from the definition of Fermi energy.

(iii) Case III: for $\epsilon = \epsilon_F$ at $T = 0K$

For $\epsilon = \epsilon_F$ and $T = 0K$

$$(\epsilon - \epsilon_F) = 0$$

Which means that numerator of $\frac{(\epsilon - \epsilon_F)}{k_B T}$ will always be $\frac{0}{0} = \text{Indeterminate}$

Therefore

$$f(\epsilon) = \frac{1}{e^{\frac{(\epsilon - \epsilon_F)}{k_B T}} + 1} = \frac{1}{e^0 + 1} = \text{Indeterminate}$$

Which means that $f(\epsilon) = \text{undetermined}$ for $\epsilon = \epsilon_F$ at $T = 0K$

Hence, the occupation of Fermi level at $T = 0 K$ has an undetermined value ranging between zero and unity (0 & 1). The Fermi-Dirac distribution function is discontinuous for $\epsilon = \epsilon_F$ at $T = 0 K$.

(iv) Case IV: for $\epsilon = \epsilon_F$ at $T > 0K$

For $\epsilon = \epsilon_F$ and $T > 0K$

$$(\epsilon - \epsilon_F) = 0$$

Which means that numerator of $\frac{(\epsilon - \epsilon_F)}{k_B T}$ will always be zero

Therefore

$$f(\epsilon) = \frac{1}{e^{\frac{(\epsilon - \epsilon_F)}{k_B T}} + 1} = \frac{1}{e^0 + 1} = \frac{1}{2}$$

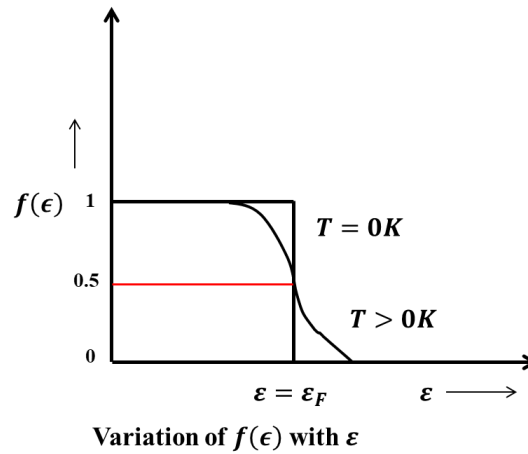
$$f(\epsilon) = \frac{1}{2}$$

Which means that $f(\epsilon) = \frac{1}{2}$ for $\epsilon = \epsilon_F$ at $T > 0K$

For $\epsilon \ll \epsilon_F$ the probability starts decreasing from 1 and reaches 0.5 at

$\epsilon = \epsilon_F$ and for $\epsilon > \epsilon_F$, it further falls off as shown in figure. In conclusion, the

Fermi energy is the most probable or average energy of the electrons in a solid.



①

Density of states

In order to know the actual number of electrons with a given energy one must know the number of states in a given system. Then multiplying the probability occupation we get the actual number of electrons. If $N(E)$ is the number of electrons in the system that have energy E and $Z(E)$ is the number of states

$$N(E)dE = Z(E)f(E)dE \quad \text{--- ①}$$

Since we know that

$$E = \frac{\pi^2 \hbar^2}{2mL^2} (n_x^2 + n_y^2 + n_z^2)$$

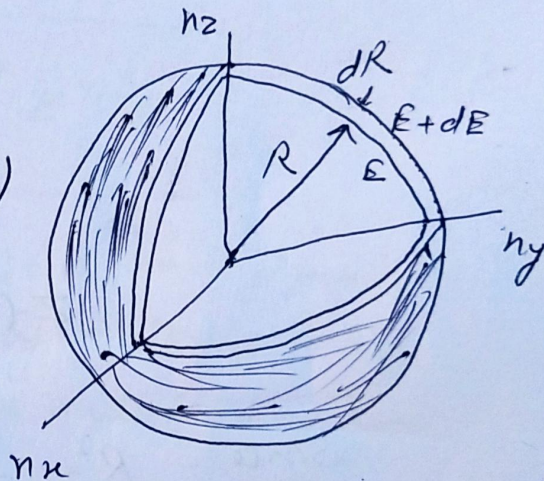
$$\text{Suppose } R^2 = n_x^2 + n_y^2 + n_z^2 \quad \text{--- ①}$$

$$\& E_0 = \frac{\pi^2 \hbar^2}{8mL^2}$$

$$E_0 = \frac{\pi^2 \hbar^2}{8\pi^2 mL^2}$$

$$E_0 = \frac{\hbar^2}{8mL^2} \quad \text{--- ②}$$

$$E = E_0 R^2 \quad \text{--- ②}$$



Equation ① represents the sphere of radius R , have energy E .

Thus the number of available states having energy E

$$= \frac{1}{8} \frac{4}{3} \pi R^3 \quad - (1)$$

Number of available states having energy $E+dE$, is

$$= \frac{1}{8} \frac{4}{3} \pi (R+dR)^3.$$

Thus the number of states having energy values lying between E & $E+dE$

$$\begin{aligned} Z(E) dE &= \frac{1}{8} \frac{4}{3} \pi [(R+dR)^3 - R^3] \\ &= \frac{1}{8} \frac{4}{3} \pi [\cancel{R^3} + \cancel{3R^2 dR} + 3R^2 dR + \cancel{3R dR^2} + \cancel{dR^3}] \\ &= \frac{1}{8} \frac{4}{3} \pi \cdot 3R^2 dR \end{aligned}$$

$$Z(E) dE = \frac{\pi}{2} R^2 dR \quad - (2)$$

Since $R^2 = \frac{E}{E_0}$

$$2R dR = \frac{1}{E_0} dE$$

$$R^2 dR = \frac{1}{2E_0} \sqrt{\frac{E}{E_0}} dE$$

$$R^2 dR = \frac{1}{2E_0^{3/2}} E^{1/2} dE$$

$$\begin{aligned} Z(E) dE &= \frac{\pi}{2} \frac{1}{2E_0^{3/2}} E^{1/2} dE \\ &= \frac{\pi}{4} \left(\frac{8mL^2}{h^2} \right)^{3/2} E^{1/2} dE \end{aligned}$$

$$Z(E) dE = \frac{\pi}{4} \frac{8 \times 2 \times 2^{1/2}}{h^3} m^{3/2} L^3 E^{1/2} dE$$

$$Z(E) dE = \frac{\pi}{4} \left(\frac{8m}{h^2} \right)^{3/2} L^3 E^{1/2} dE$$

Now from the Pauli's exclusion principle, each state permits two electrons, so that the number of energy level actually available are

$$Z(E) dE = 2 \frac{\pi}{4} \left(\frac{8m}{h^2} \right)^{3/2} L^3 E^{1/2} dE$$

Hence the number of energy states presents in a unit volume

$$g(E) dE = \frac{\pi}{2} \left(\frac{8m}{h^2} \right)^{3/2} E^{1/2} dE$$

Thus the number of electrons $N(E) dE$ in a given energy interval dE can be found by multiplying the number of energy states per unit volume multiplied by Fermi distribution function

$$N(E) dE = \frac{\pi}{2} \left(\frac{8m}{h^2} \right)^{3/2} \frac{E^{1/2}}{\left(e^{\frac{E-E_F}{k_B T}} + 1 \right)} dE$$

(1)

Effective mass of an electron: When electron is accelerated inside the crystal, the mass of an electron inside the crystal appears in general, different from free electron mass and is usually referred as effective mass m^* .

Since electrons in the crystal are referred to as quantum particles. Therefore it will have the group velocity under the influence of electric field.

$$v_g = \frac{d\omega}{dk} \quad \text{--- (1)}$$

The energy of the quantum particle is given by

$$E = \hbar \omega \quad \text{--- (2)}$$

$$\Rightarrow \omega = \frac{E}{\hbar}$$

$$\Rightarrow \frac{d\omega}{dk} = \frac{1}{\hbar} \frac{dE}{dk}$$

Therefore equation (1) can be written as.

$$dv_g = \frac{1}{\hbar} \left(\frac{dE}{dk} \right) \quad \text{--- (3)}$$

~~Acceleration due to electric field~~

Acceleration can be written as

$$a = \frac{dv_g}{dt} = \frac{1}{\hbar} \frac{d}{dt} \left(\frac{dE}{dk} \right)$$

$$a = \frac{1}{\hbar} \frac{d^2 E}{dk^2} \frac{dk}{dt} \quad \text{--- (4)}$$

②

Momentum of the quantum particle like electron
can be written as

$$p = \hbar k$$

$$\Rightarrow \frac{dp}{dt} = \hbar \frac{dk}{dt}$$

$$\Rightarrow \frac{dk}{dt} = + \frac{1}{\hbar} \frac{dp}{dt} \quad \text{--- (5)}$$

From equations (4) & (5)

$$a = \frac{1}{\hbar^2} \left(\frac{d^2 E}{dk^2} \right) \frac{dp}{dt}$$

$$\Rightarrow \frac{dp}{dt} = F_{\text{ext}} = \left(\frac{\hbar^2}{\frac{d^2 E}{dk^2}} \right) a$$

$$m^* = \frac{\hbar^2}{\frac{d^2 E}{dk^2}}$$

①

Fermi Energy : It is the energy difference between highest and lowest ~~possible~~ occupied single particle states in a quantum system of noninteracting fermions (electrons) at absolute zero. ~~(at 0K)~~ It is represented by E_F .
($T = 0^\circ K$)

Fermi Function : $\rightarrow$ This function gives the probability distribution ~~of available~~ for the available electrons energy states will be occupied at a given Temperature T K.

$$f(E) = \frac{1}{e^{\frac{E - E_F}{k_B T}} + 1}$$

E_F = Fermi energy

k_B = Boltzmann constant

T = Temperature in K.

(2)

Number of electrons per unit volume of the solid.

If n is the number of electrons per unit volume of the solid can be deduced by

$$n = \int_0^{E_F} \{(\text{Density of states in the energy range } E \text{ to } E+dE)\} \times (\text{Fermi distribution function})$$

$$n = \int_0^{E_F} g(E) dE \times f(E)$$

$$n = \int_0^{E_F} \frac{8 \pi \sqrt{2} m^{3/2}}{h^3} E^{1/2} \frac{dE}{\left(e^{\frac{E-E_F}{k_B T}} + 1 \right)}$$

Number of electrons (n) at absolute zero ($T=0^\circ\text{K}$)

& Fermi energy at ~~0K~~ $T=0^\circ\text{K}$

The number of electrons per unit volume can be written as.

$$n = \int_0^{E_F} g(E) f(E) dE \quad \text{--- (1)}$$

(3)

$$\Rightarrow n = \frac{\pi 8\sqrt{2} m^{3/2}}{h^3} \int_0^{E_F} \frac{E^{1/2} dE}{\left(e^{\frac{E-E_F}{k_B T}} + 1\right)} \quad - (2)$$

Fermi function at $T = 0^\circ K$ & $E < E_F$

$$f(E) = \frac{1}{\left(e^{\frac{E-E_F}{k_B T}} + 1\right)}$$

$$\Rightarrow \frac{(E-E_F)}{k_B T} = \frac{(E-E_F)}{k_B \times 0} = \frac{-ive}{0} = -\infty$$

$\rightarrow E-E_F = (\text{fine})$
for $E < E_F$

$$\Rightarrow f(E) = \frac{1}{e^{-\infty} + 1} = \frac{1}{\frac{1}{\infty} + 1} = 1$$

Now the equation (2) can be written as.

$$n = \frac{\pi 8\sqrt{2} m^{3/2}}{h^3} \int_0^{E_F} E^{1/2} dE$$

$$= \frac{\pi 8\sqrt{2} m^{3/2}}{h^3} \cdot \frac{2}{3} E_F^{3/2}$$

~~8~~ -
 $8 \times 2 \times \sqrt{2}$
 $= 8^{3/2}$

$$\boxed{n = \frac{\pi}{3} \left(\frac{8m}{h^2} \right)^{3/2} E_F^{3/2}} \quad \text{at } T = 0^\circ K$$

$$\boxed{E_F = \left(\frac{h^2}{8m} \right) \left(\frac{3}{\pi} \right)^{2/3} n^{2/3}} \quad \text{at } T = 0^\circ K$$

$E_F = 5.85 \times 10^{-38} \cdot n^{2/3}$ where $B = \frac{h^2}{8m} \left(\frac{3}{\pi} \right)^{2/3}$ Joules

$$E_F = B n^{2/3}$$

~~$$B = \frac{h^2}{8m} \left(\frac{3}{\pi} \right)^{2/3}$$~~

$$B = \frac{h^2}{8m} \left(\frac{3}{\pi} \right)^{2/3}$$

$$B = 5.85 \times 10^{-38} \text{ Joules}$$

for electron.

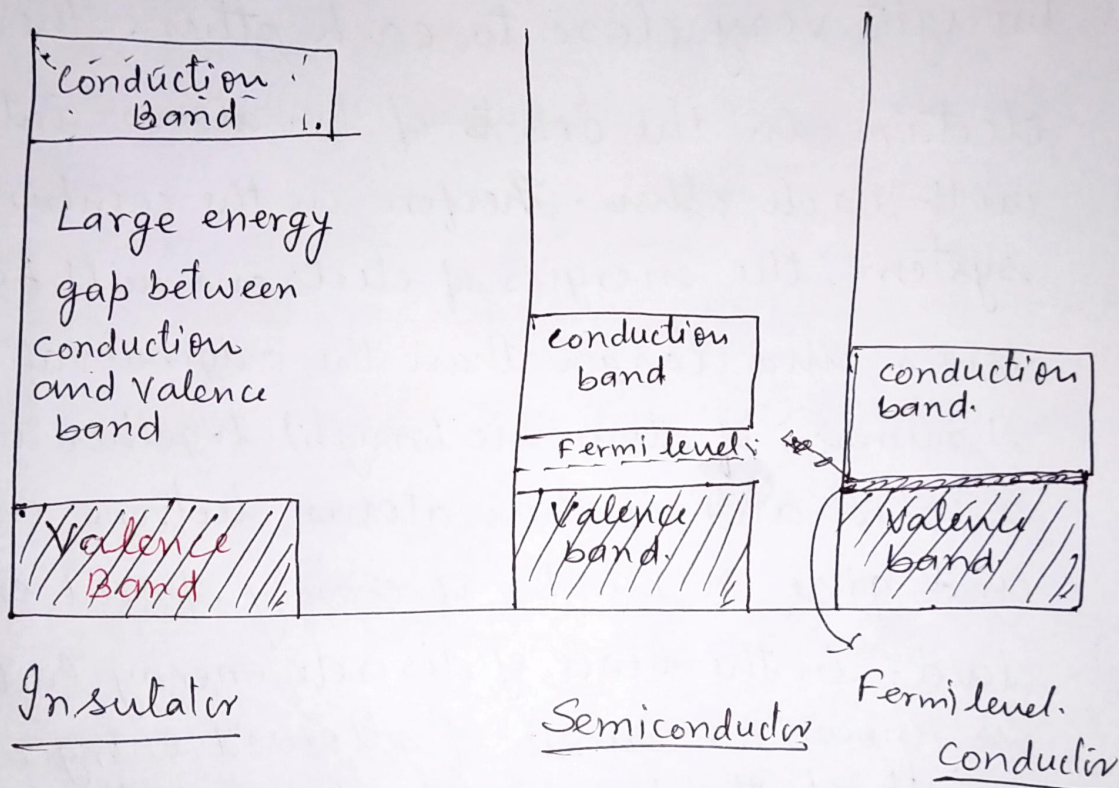
Origin of Energy bands

Since we know that isolated atoms have discrete energy levels. Suppose two isolated atoms are brought very close to each other. Then the electrons in the orbits of two atoms interact with each other. Therefore in the combined system, the energies of electrons will be slightly lower and larger than the original value. If N number of atoms are brought together to form a solid and if these atoms electrons interact and give N number of closely spaced energy levels in the place of discrete energy levels, it is known as bands of allowed energies. The width of the bands depends upon the degree of overlapping of electrons of adjacent atoms and is largest for outermost electrons.

The band corresponding to outermost orbits is called conduction band and next inner band is called valance band. The gap between these two allowed bands is called forbidden energy gap or band gap.

classification of crystalline solids

Based on the width of forbidden gap (band) solids are classified into insulators, semiconductors and conductors.



- (i) conductors $\rightarrow$ The Valence band and conduction bands overlapped with each other and the energy gap E_g is zero. At room temperature free electrons already exists in huge number at conduction band. Hence these solids are good electrical conductors as well as good thermal conductors. The electrical resistivity increases at high temperature by collisions among the free electrons
- Ex Au, Ag Cu, Al etc.

③ ⑥

Semiconductor $\rightarrow$ The valence band and conduction bands are separated with a small energy band gap $E_g \approx 1.0 \text{ eV}$. At 0 K temperature, free electrons are not available in conduction band and hence they behave like insulators. The electrical conductivity increases at ~~high~~ temperatures by transition of free electrons from valence band to conduction band. Thus these solids behave like conductors at high temperatures.

Ex: Silicon $E_g = 1.1 \text{ eV}$, Germanium $E_g = 0.7 \text{ eV}$

Insulators $\rightarrow$ Valence band and conduction bands are separated by a very large energy gap $E_g > 3 \text{ eV}$. At room temperatures the conduction band is empty and valence band is full of electrons. Hence these solids are electrical insulators. Even at high temperatures valence electrons are unable to jump into the conduction band.

Ex. Glass, Mica, Rubber etc.

Theory of Semiconductor

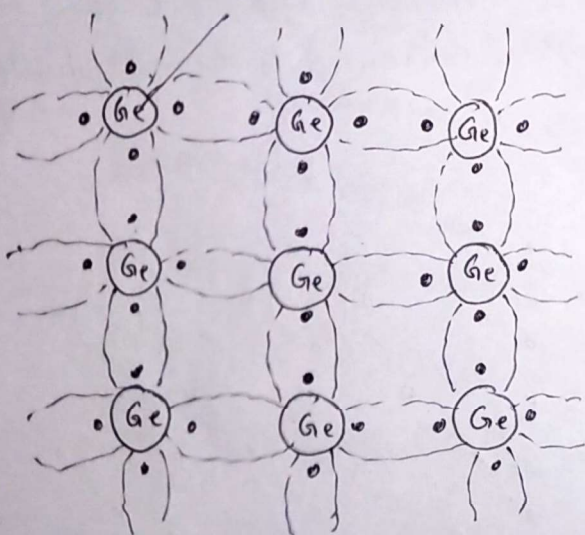
- There are two kind of semiconductors.
- Pure semiconductors are called intrinsic semiconductors.
 - Impure semiconductors are called extrinsic semiconductors.

Intrinsic Semiconductors (Ex Si & Ge)

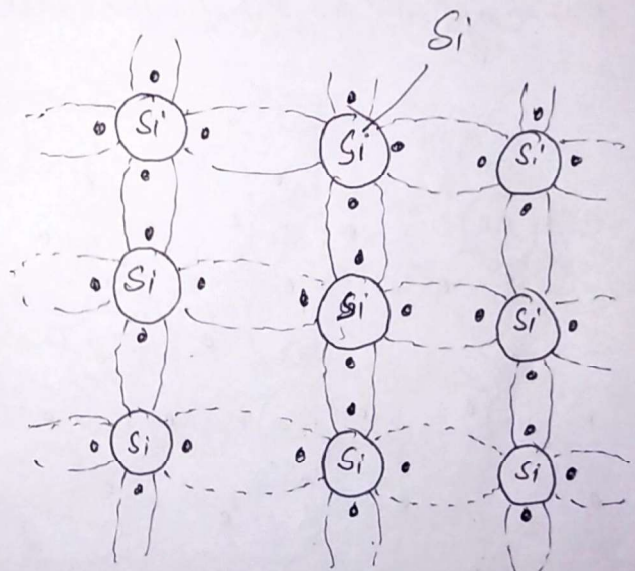
Si (Silicon) $14 - 1s^2 2s^2 2p^6 3s^2 3p^2 \rightarrow 4$ valence electrons

Ge (Germanium) $32 - 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^2 \rightarrow 4$ valence electrons

The semiconductors like Ge & Si in its pure form are called intrinsic semiconductors.



Germanium



Silicon

Fig ①

Germanium and Silicon are two most important semiconductors used in electronic devices. The Germanium (Ge) and Silicon (Si) has four valence

(2)

electrons. Each Ge & Si atoms form a covalent bond with its neighbouring atoms. The symbolic structure of Ge & Si is shown in the fig ①.

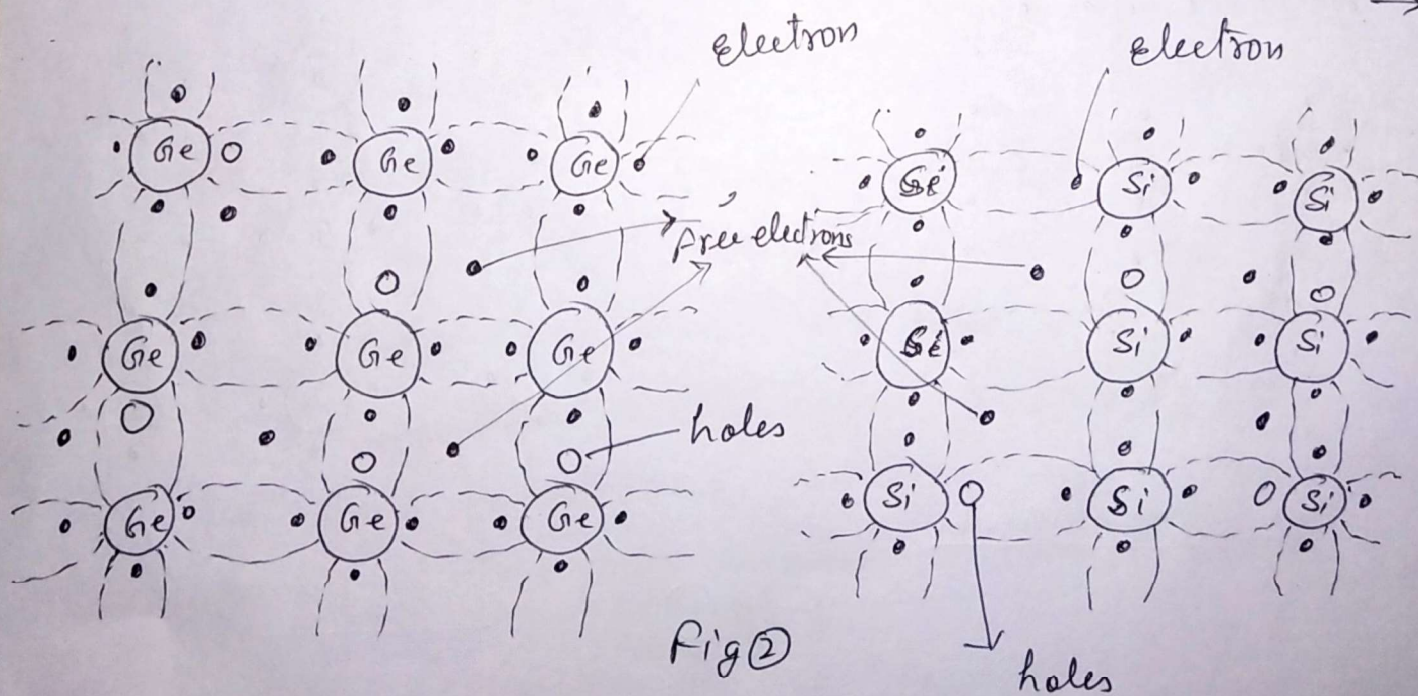
Since all valence electrons are bound with neighbouring atoms and no free electrons are available, at

~~low at low temperatures. As soon as the temperature~~
~~increases and reached at room temperature the ideal~~
~~structure~~

~~low temperature (0°K) ⇒~~

⇒ and the Ge & Si crystals behaves like insulators.

However at room temperature, some of the covalent bonds gets broken because of the thermal energy applied to the crystal (as the temperature of the crystal enhanced to room temperature) ⇒



→ some of the covalent bonds broken up

(3)

and conduction is made possible as the electrons (very few in numbers) free to move in the crystal.

The energy E_g required to break the covalent bonds in Ge is $E_{(Ge)g} = 0.72 \text{ eV}$ and $E_{(Si)g} = 1.1 \text{ eV}$ for Si.

The absence of electrons in the covalent bonds is shown in the Fig (2) by a small circle and such an incomplete covalent bond is called hole

Mechanism by which hole contribute to the conductivity

⇒ When an electron exit from the covalent bond a hole (deficiency of electron) ~~exists~~ exists.

It is relatively easy for a valence electron in the neighbouring atom to leave its covalent bond and to fill the hole. An electron moving from a bond to fill a hole leaves a hole in its initial position. Hence the hole effectively moves in the direction opposite to that of electrons. This hole in its new position may now be filled by an electron from another covalent bond

⇒ In pure or intrinsic semi conductors the number of holes is equal to number of electrons.

$$n = p = n_i \quad \text{--- (1)}$$

where n = number density of electrons, p = number density of holes
 n_i = intrinsic carrier concentration

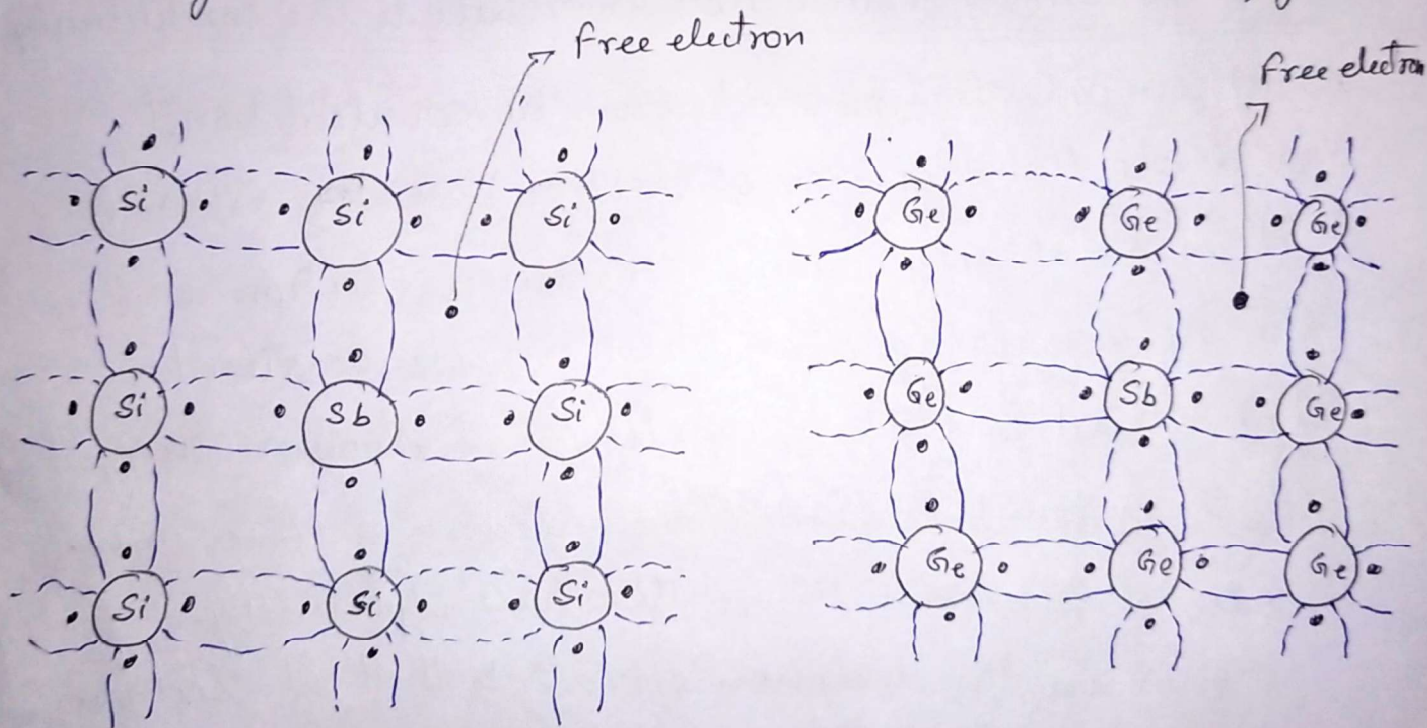
Doner & Acceptor Impurities :-

Doners :- $\rightarrow$ If a small percentage of trivalent or pentavalent atoms is added to the Ge, or Si (intrinsic semiconductor), a doped or extrinsic semiconductor is formed.

If the dopant has five valance electrons (Sb)

Antimony (Sb) $\{A=51\}$ $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^3$

The crystal structure will become as shown in the figure



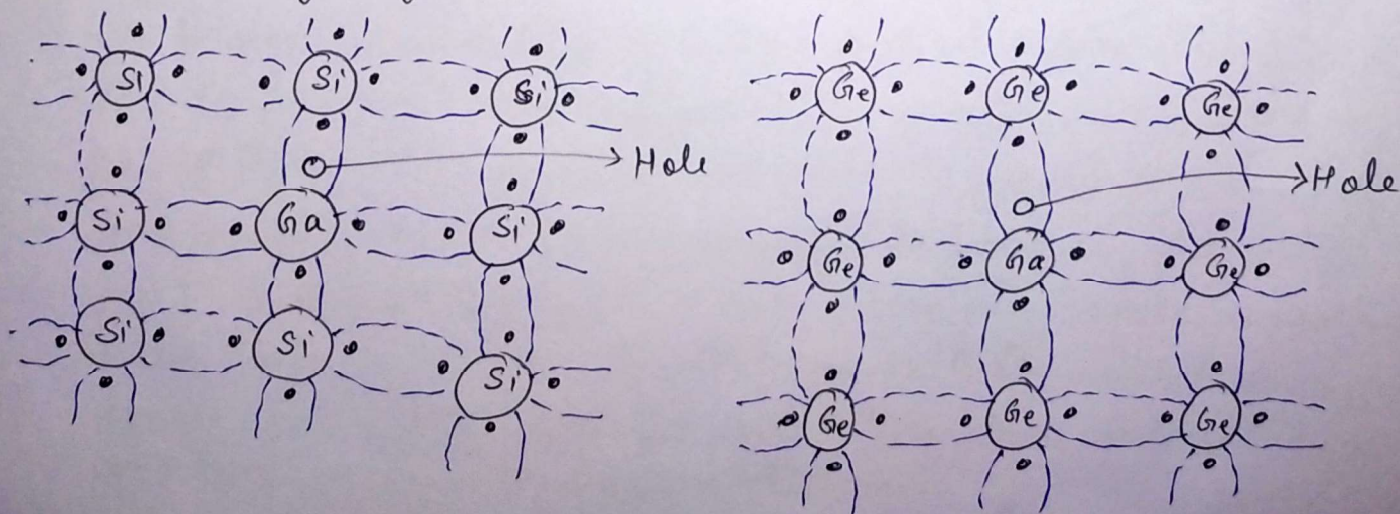
The four of the five ~~electrons~~ valence electrons will ~~occupy~~ occupy covalent ~~bonds~~ bonds and fifth will be available for the carrier current. The energy required to detach the fifth electron from the atom is

of the order of 0.01 eV for Ge & 0.05 eV for Si.
Suitable pentavalent impurities are, Antimony (Sb)
Arsenic (As) & Phosphorus (P)

If the impurities added to the intrinsic semiconductor (Ge or Si) is having 5th valency (pentavalent like Sb, As. etc) then the fifth electron of the dopant left over as free. Such type of impurities which provides the extra electrons for the conduction is called donors.

Acceptors $\rightarrow$ If the trivalent impurity like ~~Boron~~ Boron, Gallium (Ga), Indium In.

Acceptors $\rightarrow$ If the trivalent impurity like ~~Boron~~ Boron (B) Gallium (Ga) or Indium (In) is added to the intrinsic semiconductor, only three of the covalent bonds can be filled, and the vacancy that exists in the fourth bond constitutes a hole. This situation is illustrated in the following figure.



such impurities make available positive carriers named Holes, which accept electrons. These impurities are consequently known as acceptors.

n-type Semiconductor $\Rightarrow$ If donor impurity is ~~having~~ pentavalent (+5), and added to the pure Ge or Si, negative charge carriers becomes more than positive charge carrier. These type of semiconductor is called n-type semiconductor.

p-type semiconductor $\Rightarrow$ If acceptor impurity is trivalent, and is added to the pure Ge or Si, then the positive charge carriers (Holes) are more than negative charge carriers, then these types of semiconductors are called p-type semiconductors.

Fermi level in an intrinsic semiconductor

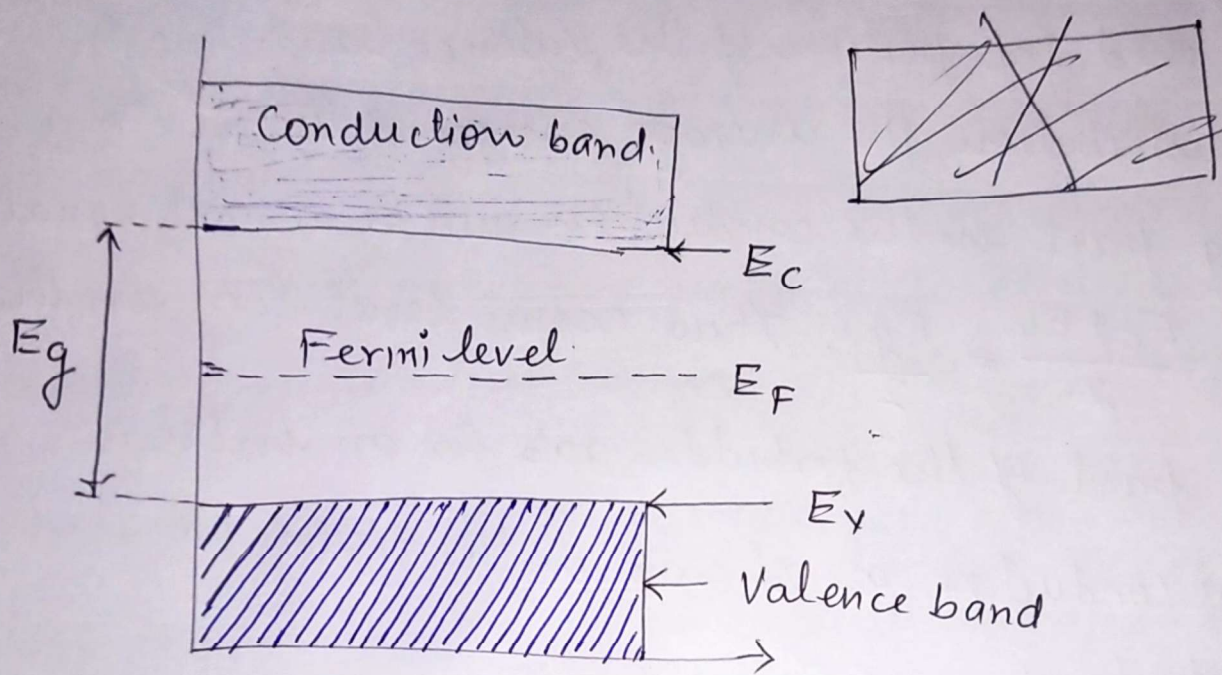


Fig (1)

Energy gap E_g is the difference between the bottom of the conduction band (E_C) and top of the valence band (E_V) as shown in the above figure. $E_g = E_C - E_V$ Conventionally energy at the top of the valence band (~~E_V~~) is taken as zero ($E_V = 0$) for reference.

At $T = 0\text{ K}$, energy levels in the valence band are completely filled and all energy levels in the conduction band are completely empty. But at room temperature due to thermal excitation some of the electrons at the top of the valence band are able to jump the energy gap and occupy some energy levels at the bottom part of the conduction band. These electrons return soon to the valence band. The electrons in these set of energy levels continue to undergo excitation and deexcitation

Therefore the $\propto$ conduction electrons between the energy levels in the bottom part of the conduction band and top portion of the valence band. Due to this distribution, the average energy of the electrons taking part in the conduction will be almost equal to $\frac{E_c + E_v}{2} = \frac{E_g}{2}$. Thus Fermi level lies in the mid part of the forbidden gap for an intrinsic semiconductor as shown in the figure (1)

Impurity levels and Fermi levels in an extrinsic semiconductor

(i) n-type semiconductor $\rightarrow$

In n-type semiconductor, electrons of donor atoms are free to move and hence they possess more energy. As a result the energy level for the electrons gets elevated to a position much higher than the valence band and lie very close to the conduction band as shown in the figure (2).

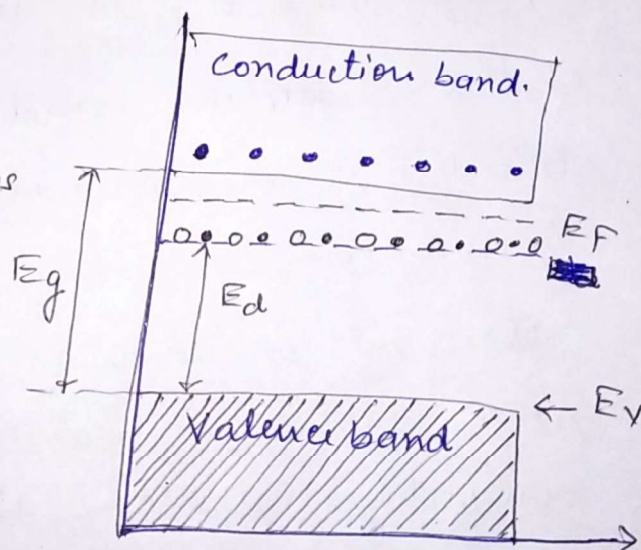


Fig (2) Fermi level

Fig (2) Fermi level in n-type semiconductor.

The energy difference between top of the valence band and the donor level is denoted by E_d . The electrons present in the donor level are responsible for the ~~re~~ conduction in n-type semiconductors.

Since electrons present in the donor levels stay very close to the bottom of the conduction band, therefore these electrons get distributed between energy levels in the bottom part

of the conduction band and the donor levels. The difference between these two levels, ~~$(E_c + E_d)$ or $(E_g + E_d)$~~ as shown in figure ② $\rightarrow (E_c - E_d)$ or $(E_g - E_d)$

$\Rightarrow$

Therefore the average energy of the electrons participating in conduction is $\frac{E_c + E_d}{2}$ or $\frac{(E_g + E_d)}{2}$

Hence the Fermi level of n-type semiconductor at low temperatures will be located in the forbidden band at the level $\frac{(E_c + E_d)}{2}$ or $\frac{(E_g - E_d)}{2}$ just below the bottom of the conduction band.

(ii) p-type semiconductors $\rightarrow$

In p-type semiconductors acceptor atoms (trivalent impurity) gives rise to holes which are free to move in the conduction band.

Since the energy of the holes increases opposite to the electrons and hence they occupy energy levels in the band gap close to the valence band as shown in

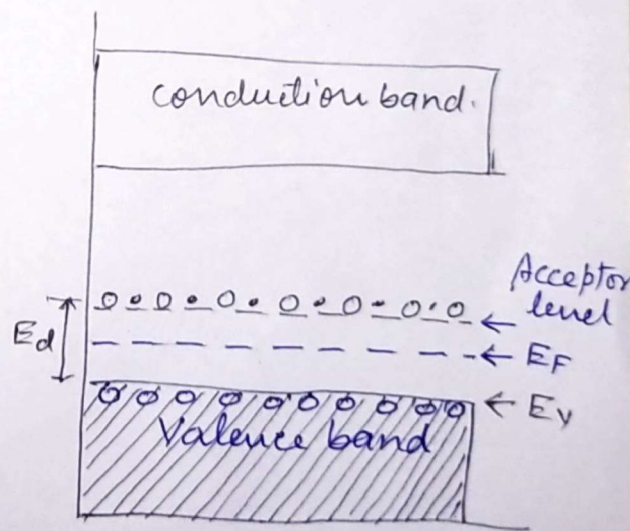


Fig ③ Fermi level in p-type semiconductor.

figure ③. The energy difference between acceptor levels and top of the valence band is denoted by $E_a - E_v = E_a$ as $E_v = 0$. At low temperatures the holes participate in the conduction undergo excitation and de-excitation between acceptor levels

and energy levels in the top portion of valence band.

As a result the average energy of the holes participating in conduction becomes $\frac{E_a + E_v}{2}$ or $E_g/2$

Hence the Fermi level in p-type semiconductors at low temperatures will be located in the forbidden gap at the level $\frac{E_a + E_v}{2}$ or $\frac{E_g}{2}$ from the top of the valence band.

Carrier concentration in an intrinsic semiconductor ①

The number of electrons in the conduction band/unit volume of the material is called the electron concentration or electron density. Similarly the number of holes in the valence band/unit volume of the material is called the hole concentration or hole density.

The number of electrons per unit volume in the energy range E & $E+dE$

$$dn_e = g(E) dE f(E) \quad - (1)$$

Where $g(E) dE$ is the number of allowed energy states per unit volume in the energy range E & $E+dE$ and $f(E)$ is fermi distribution function, represents the probability of occupation of states with energy E .

The concentration of electrons in the conduction band (n_e) can be obtained by integrating eq (1) from ~~E_c~~ to $E = E_c$ to $E = \infty$

$$n_e = \int_{E_c}^{\infty} g(E) dE \times f(E) \quad - (2)$$

$$n_e = \frac{\pi}{2} \left(\frac{8 m_e}{h^2} \right)^{3/2} \int_{E_c}^{\infty} \frac{E^{1/2}}{e^{\frac{E-E_F}{k_B T}} + 1} dE \quad - (3)$$

In the ~~po~~ semiconductors (intrinsic) the electrons are not free but it is moving in the periodic potential so m_e should be replaced by m_e^* , the effective mass of the electrons.

Further in the semiconductors, the energy of the electrons starts from E_c , hence the energy of the electron E is replaced by $(E - E_c)$. Therefore equation (3)

can be written as

$$n_e = \frac{\pi}{2} \left(\frac{8m_e^*}{h^2} \right)^{3/2} \int_{E_c}^{\infty} \frac{(E - E_c)^{1/2} dE}{\left(e^{\frac{E - E_F}{k_B T}} + 1 \right)} \quad - (4)$$

Since the Fermi ~~level~~ energy is within the forbidden energy band gap. For electrons in the conduction band $E > E_c$; ~~so~~ ~~we~~ If $(E_c - E_F) \gg k_B T$, then

$$E - E_F \gg k_B T$$

$$\Rightarrow \frac{(E - E_F)}{k_B T} \gg 1$$

$$\Rightarrow e^{\frac{(E - E_F)}{k_B T}} + 1 \approx e^{\frac{(E - E_F)}{k_B T}} \quad - (5)$$

Now substituting the value of equation (5) in (4) we have

$$n_e = \frac{\pi}{2} \left(\frac{8m_e^*}{h^2} \right)^{3/2} \int_{E_c}^{\infty} (E - E_c)^{1/2} e^{-\frac{(E - E_F)}{k_B T}} dE \quad - (6)$$

Multiplying and dividing equation (6) by $(k_B T)^{3/2}$

$$n_e = \frac{\pi}{2} \left(\frac{8m_e^*}{h^2} \right)^{3/2} (k_B T)^{3/2} \int_{E_c}^{\infty} \left(\frac{E - E_c}{k_B T} \right)^{1/2} e^{-\frac{(E - E_c)}{k_B T}} \times e^{-\frac{(E_c - E_F)}{k_B T}} \frac{dE}{k_B T}$$

$$\Rightarrow n_e = \frac{\pi}{2} \left(\frac{8m_e^*}{h^2} \right)^{3/2} (k_B T)^{3/2} e^{-\frac{(E_c - E_F)}{k_B T}} \times \int_{E_c}^{\infty} \frac{(E - E_c)^{1/2}}{(k_B T)^{1/2}} e^{-\frac{(E - E_c)}{k_B T}} \frac{dE}{k_B T} \quad (7)$$

substituting

$$\frac{E - E_c}{k_B T} = x$$

$$\frac{dE}{k_B T} = dx$$

E	x
E_c	0
∞	∞

$$n_e = \frac{\pi}{2} \left(\frac{8m_e^*}{h^2} \right)^{3/2} (k_B T)^{3/2} \int_0^{\infty} x^{1/2} e^{-x} dx \times e^{-\frac{(E_c - E_F)}{k_B T}} \quad (8)$$

$$\int_0^{\infty} x^{1/2} e^{-x} dx = \frac{\sqrt{\pi}}{2}$$

$$n_e = \frac{\pi}{2} \frac{\sqrt{\pi}}{2} \left(\frac{8m_e^*}{h^2} \right)^{3/2} (k_B T)^{3/2} \times e^{-\frac{(E_c - E_F)}{k_B T}}$$

$$\Rightarrow n_e = \frac{\pi^{3/2}}{4} \frac{2}{8} 2^{3/2} \left(\frac{m_e^* k_B T}{h^2} \right)^{3/2} e^{-\frac{(E_c - E_f)}{k_B T}} \quad (4)$$

$$\Rightarrow n_e = 2 \left(\frac{2 \pi m_e^* k_B T}{h^2} \right)^{3/2} e^{-\frac{E_c - E_f}{k_B T}} \quad (8)$$

$$\text{Suppose } N_c = 2 \left(\frac{2 \pi m_e^* k_B T}{h^2} \right)^{3/2}$$

$$n_e = N_c e^{-\frac{E_c - E_f}{k_B T}} \quad (9)$$

Equations (8) & (9) represents the expression for electron concentration in conduction band of intrinsic semiconductor. Where N_c is temperature dependent material constant known as effective density of states in the conduction band.

Hole concentration in intrinsic semiconductor (5)

Number of holes per unit volume in a semiconductor in the energy range E & $E+dE$.

$$dN_h = f_h(E) g_h(E) dE \quad \text{--- (1)}$$

Where $g_h(E) dE$ is the density of states available for holes in the energy range E & $E+dE$ in the valence band and $f_h(E)$ is the probability of finding the holes in the energy state E .

Since the probability of having hole at some energy E in the valence band is just the same as the probability of absence of an electron at the same energy E .

Probability of absence = 1 - probability of presence

Hole probability = 1 - electron probability.

$$\Rightarrow f_h(E) = 1 - \frac{1}{\left(e^{\frac{E-E_F}{k_B T}} + 1 \right)} = \frac{e^{\frac{E-E_F}{k_B T}}}{\left(e^{\frac{E-E_F}{k_B T}} + 1 \right)}$$

$$f_h(E) = \frac{1}{1 + e^{-\frac{(E-E_F)}{k_B T}}}$$

(6)

Since $k_B T \ll \phi$

$$e^{-\frac{(E-E_F)}{k_B T}} = e^{\frac{(E_F-E)}{k_B T}} \gg 1$$

$$\Rightarrow e^{\frac{E_F-E}{k_B T}} + 1 \approx e^{\frac{(E_F-E)}{k_B T}}$$

$$f_h(E) = e^{-\frac{(E_F-E)}{k_B T}} = e^{\frac{E-E_F}{k_B T}}$$

$$\Rightarrow dn_h = \frac{\pi}{2} \left(\frac{8m_h^*}{h^2} \right)^{3/2} \frac{(E_V-E)^{1/2} dE}{e^{\frac{E-E_F}{k_B T}}}$$

$$n_h = \frac{\pi}{2} \left(\frac{8m_h^*}{h^2} \right)^{3/2} \int_{-\infty}^{E_V} (E_V-E)^{1/2} e^{\frac{E-E_F}{k_B T}} dE$$

$$= \frac{\pi}{2} \left(\frac{8m_h^*}{h^2} \right)^{3/2} \frac{1}{(k_B T)^{3/2}} \int_{-\infty}^{E_V} \frac{(E_V-E)^{1/2}}{(k_B T)^{1/2}} e^{-\frac{(E_V-E)}{k_B T}} e^{\frac{(E_V-E_F)}{k_B T}} \frac{dE}{k_B T}$$

$$= \frac{\pi}{2} \left(\frac{8m_h^*}{h^2} \right)^{3/2} e^{\frac{(E_V-E_F)}{k_B T}} \int_0^{\infty} x^{1/2} e^{-x} dx$$

Where $x = \frac{E_V-E}{k_B T}$

$$n_h = 2 \left(\frac{2m_h^* k_B T}{h^2} \right)^{3/2} e^{-\frac{(E_F-E_V)}{k_B T}} = N_V e^{-\frac{(E_F-E_V)}{k_B T}} \quad - (2)$$

(7)

Relation between Fermi energy and energy gap
for an intrinsic semiconductor:-

OR Fermi energy in intrinsic semiconductor

For an intrinsic semiconductor, the number of electrons per unit volume in conduction band is equal to the number of holes per unit volume in valence band.

$$n_e = n_h \quad \text{--- (1)}$$

$$\Rightarrow 2 \left(\frac{2 m_e^* k_B T}{h^2} \right)^{3/2} e^{-\frac{(E_c - E_F)}{k_B T}} = 2 \left(\frac{2 m_h^* k_B T}{h^2} \right)^{3/2} e^{+\frac{(E_v - E_F)}{k_B T}}$$

$$\Rightarrow (m_e^*)^{3/2} e^{-\frac{(E_c - E_F)}{k_B T}} = (m_h^*)^{3/2} e^{+\frac{(E_v - E_F)}{k_B T}}$$

$$e^{\frac{(-E_c + E_F - E_v + E_F)}{k_B T}} = \left(\frac{m_h^*}{m_e^*} \right)^{3/2}$$

$$2 E_F - (E_c + E_v) = \frac{3}{2} k_B \log_e \left(\frac{m_h^*}{m_e^*} \right)^{3/2}$$

$$E_F = \left(\frac{E_c + E_v}{2} \right) + \frac{3}{4} k_B \log_e \left(\frac{m_h^*}{m_e^*} \right)$$

(8)

$$E_F = \frac{E_C + E_V}{2} + \frac{3}{4} k_B T \log \left(\frac{m_h^*}{m_e^*} \right)$$

At absolute zero i.e. $T = 0 \text{ K}$

$$E_F = \frac{E_C + E_V}{2}$$

Since $E_V = 0$ & $E_C = E_g$

$$E_F = E_g / 2$$

Intrinsic carrier concentration

In an intrinsic carrier semiconductors there will be continuous production of electron hole pairs due to thermal energy. At the same time there will be decrease in numbers of electrons and holes due to electron-hole recombination. As long as temperature remains constant the electron hole concentration will also remain constant. Therefore at temperature $T \text{ K}$ $n_e = n_h = n_i$

Therefore the product of concentration
 $n_i^2 = n_e n_h$ — ①

(9)

$$n_i^2 = N_c N_v e^{-\frac{(E_c - E_f)}{k_B T}} e^{-\frac{(E_f - E_v)}{k_B T}}$$

$$n_i^2 = N_c N_v e^{-\frac{(E_c - E_v)}{k_B T}}$$

$$\Rightarrow n_i^2 = N_c N_v e^{-\frac{E_g}{k_B T}}$$

$$\Rightarrow n_i = \sqrt{N_c N_v} e^{-\frac{E_g}{2 k_B T}}$$

$$\Rightarrow n_i = \left[2 \left(\frac{2\pi m_e^* k_B T}{h^2} \right)^{3/2} \times 2 \left(\frac{2\pi m_h^* k_B T}{h^2} \right)^{3/2} \right]^{1/2} e^{-\frac{E_g}{2 k_B T}}$$

$$\Rightarrow n_i = 2 \left(\frac{2\pi k_B T}{h^2} \right)^{3/2} (m_e^* m_h^*)^{3/4} e^{-E_g/k_B T} \quad \text{--- (1)}$$

$$\Rightarrow n_i = C T^{3/2} e^{-E_g/2 k_B T} \quad \text{--- (2)}$$

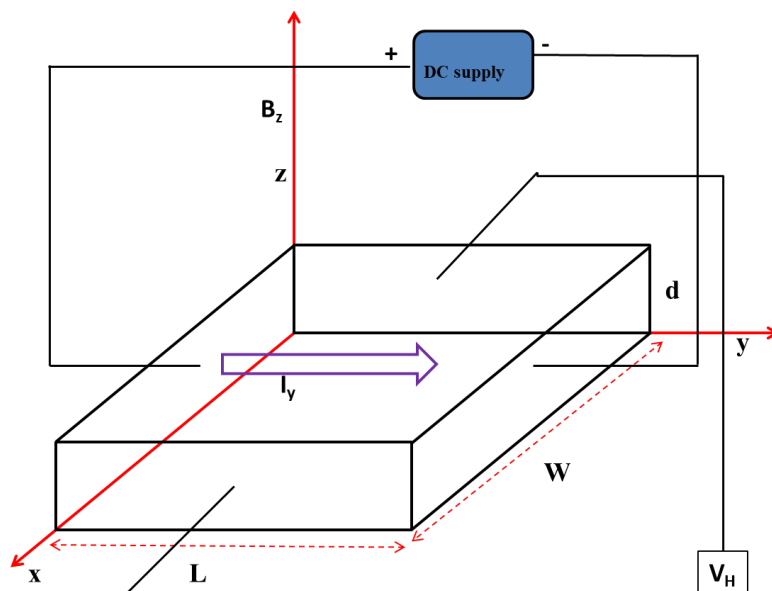
$$\text{where } C = 2 \left(\frac{2\pi k_B}{h^2} \right)^{3/2} (m_e^* m_h^*)^{3/4}$$

Equation (1) & (2) represents the charge carrier concentrations in an intrinsic semiconductor.

Hall Effect:

The Hall Effect is a phenomenon where a solid (metal or Semiconductor) material, conducting an electric current, generates a transverse electric field when exposed to a magnetic field perpendicular to the current.

Suppose an electric an electric field E is applied across the two faces of the semiconductor slab (DC Supply). The slab length is L width W and thickness d . Semiconductor slab is placed in an external magnetic field B_z , which is along z direction. When the slab is connected through the DC power supply a current I , is start flowing across the semiconductor slab as shown in the Figure.



Since current flows due to motion of the electrons/holes and these electrons will experience magnetic force $F = e v_y B_z \dots \dots \dots (1)$

Due to this magnetic force, electrons starts accumulating at one surface and an induced electric field starts developed resulting a Hall voltage V_H .

When the electric force due to developed Hall electric field $E_H = \frac{V_H}{W} \dots \dots \dots (2)$ becomes equal to magnetic force given in equation (1).

Then $e E_y = e E_H = e v_y B_z$

$$E_H = v_y B_z \dots \dots \dots (2)$$

$$V_H = W v_y B_z \dots \dots \dots (3)$$

Since $v_y = v_d$ (drift velocity)

For n-type semiconductor

Current density $I_y = e n_e v_d A = e n_e v_d W d$

Where $A = W d$ refer to above figure

Using above equations we get

$$\text{Hall voltage } V_H = \frac{I_y B_z}{e n_e d} \dots \dots \dots (4)$$

Hall Effect sensors: It is a kind of magnetic sensor which can be used for detecting the strength and direction of magnetic field produced from the permanent magnet or an electromagnet.

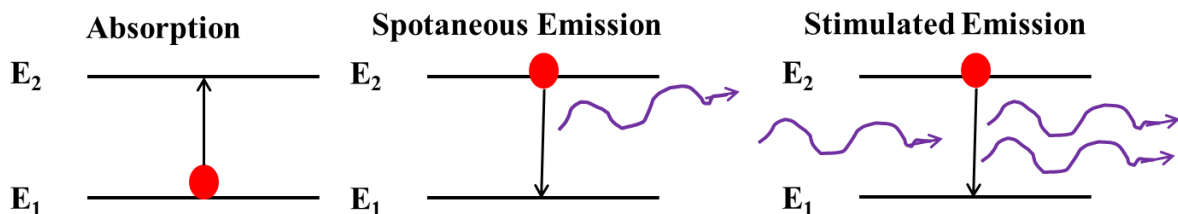
Magnetic Switches: Magnetic switch consists of flat contact plates sealed in a glass tube filled with an inert gas. When a permanent magnet approaches, the contact plates begin to attract each other and close. When the permanent magnet is removed, the contact plates instantly demagnetize and return to their original position.

Magnetic switches excel in challenging environments, effortlessly handling dirt, humidity, gas, dust, chips, and more. With a stable switching point and a remarkable reproducibility error of just 0.01 mm, they guarantee reliable performance.

Absorption: When an atom situated in a lower energy level absorbs a photon with a frequency of $h\nu$, it transitions to a higher energy level is call absorption.

Spontaneous emission: An atom positioned in a higher energy level can spontaneously transition to a lower level, emitting a photon with a frequency of $h\nu$ if the transition between E_2 and E_1 is radiative. This emitted photon has a random direction and phase. Such transition is called spontaneous emission.

Stimulated emission: An incoming photon prompts an atom in an upper energy level to decay, releasing a "stimulated" photon that shares identical properties with the incident photon. The term "stimulated" emphasizes that this radiation occurs only in the presence of an incident photon. The amplification results from the similarities between the incident and emitted photons.



Optical Loss and Gains:

Optical losses and gains refer to the changes in optical power or intensity experienced by light as it travels through a medium or an optical system. These terms are fundamental in understanding the behaviour of light in various contexts, such as in optical fibres, lasers, or other optical devices.

Optical Losses:

1. Absorption:

- Occurs when light is absorbed by a material rather than transmitted through it. The absorbed energy is converted into other forms, typically heat, reducing the transmitted light's intensity.

2. Scattering:

- Light gets redirected or scattered due to irregularities or impurities in the medium. Scattering causes light to deviate from its original path, leading to a loss of intensity or coherence.

3. Reflection:

- When light encounters a boundary between two materials, some of it reflects back into the original medium. Reflection at interfaces can cause losses in optical systems.

4. Bend Loss:

- In optical fibres or waveguides, light can be lost due to bending or curvature, causing some of the light to escape the fibre core, reducing transmitted intensity.

Optical Gains:

1. Stimulated Emission:

- The process where incoming photons cause excited atoms or molecules to emit additional photons of the same frequency and phase. This amplifies the light, leading to optical gain.

2. Population Inversion:

- Achieving a state where more atoms or molecules are in an excited state than in the ground state. This condition is essential for stimulated emission and achieving optical gain in laser systems.

3. Amplification:

- Optical gain occurs when the intensity of light increases as it passes through a gain medium due to stimulated emission dominating over absorption.

Understanding and managing optical losses and gains are critical in designing and optimizing optical systems and devices, such as lasers, optical amplifiers, and communication systems. Minimizing losses and maximizing gains are essential for efficient and high-performance optical devices and applications.

Application of lasers in security, holography and medical fields

Lasers have diverse applications across various fields due to their focused, coherent, and intense nature. Here's how they're utilized in security, holography, and the medical field:

Security:

1. Laser Security Systems:

- Lasers are employed in security systems to create intricate laser grids or barriers. Intrusion detection is triggered when the grid is breached, alerting security personnel.

2. Laser Sensors:

- Laser-based sensors detect movement or disruptions in sensitive areas. They're used in perimeter defense systems and can be highly accurate in detecting unauthorized access.

Holography:

1. Holographic Security Features:

- Lasers are instrumental in creating intricate holograms used in security features for IDs, credit cards, or valuable products. These holograms are difficult to replicate, enhancing security.

2. 3D Imaging and Displays:

- Lasers enable the creation of high-quality, three-dimensional holographic images and displays. They find applications in entertainment, art, and advanced visualization.

Medical Field:

1. Surgery and Medical Treatments:

- Lasers are widely used in various medical procedures, such as laser surgery for precise incisions, tissue ablation, and treatments for eye conditions like LASIK.

2. Diagnostic Techniques:

- Laser-based diagnostic tools like laser-induced breakdown spectroscopy (LIBS) or laser imaging methods aid in detecting diseases, analyzing tissues, and performing non-invasive imaging.

3. Therapeutic Applications:

- Laser therapy is used in physiotherapy and pain management. Low-level laser therapy stimulates tissue repair and provides relief from chronic pain.

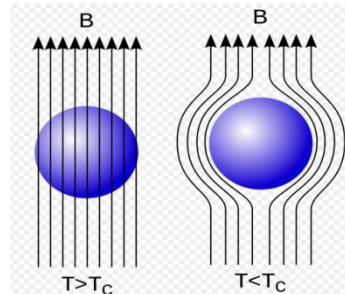
4. Dermatology and Aesthetics:

- Lasers are utilized in dermatology for tattoo removal, hair removal, skin resurfacing, and addressing vascular lesions or pigmentation disorders.

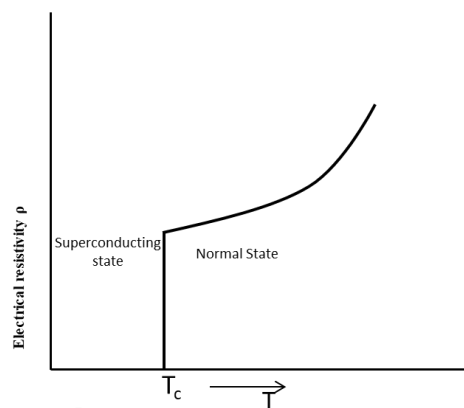
Lasers' versatility and precision make them invaluable across security systems, holography for authentication, and a wide array of applications within the medical field, revolutionizing treatments, diagnostics, and therapeutic procedures.

- 1. Meissner Effect:** When a material makes the transition from the normal to superconducting state, it excludes magnetic fields from its interior; this is called the Meissner effect.

Or when materials, are cooled below the critical temperature, they expel magnetic field and do not allow the magnetic field to penetrate inside them. This phenomenon in superconductors is called Meissner effect.



Critical temperature T_c of superconductor: Temperature at which a normal material abruptly changes into the superconductor is called the transition or critical temperature.



Superconductor and Superconductivity:

A superconductor is a material that, when cooled below a certain critical temperature, exhibits remarkable properties, notably the complete absence of electrical resistance and the expulsion of magnetic field (the Meissner effect). ***This property is called superconductivity.*** *In a superconducting state, electrical currents can flow indefinitely without experiencing any loss of energy due to resistance.*

Type I Super conductors: Type I superconductors' exhibits complete Meissner effect. At critical field magnetisation drops to zero. Have only one critical field. Transition at critical magnetic field from superconductor to normal conductor is reversible.

Type II Super conductors: These superconductors have two critical fields, and transition from superconductor to normal conductor occurs gradually as the magnetic field increased. They have intermediate state between the critical fields and are called mixture of the normal and super conducting states.

BCS Theory

The BCS theory, developed by John Bardeen, Leon Cooper, and Robert Schrieffer, is a fundamental theory in condensed matter physics that explains superconductivity in certain materials at low temperatures.

Key Aspects of the BCS Theory:

1. **Electron Pairing Mechanism:**
 - **Cooper Pairs:** *BCS theory proposes that at very low temperatures, electrons in a metal form pairs, known as Cooper pairs, due to interactions with lattice vibrations (phonons).*
2. **Zero Electrical Resistance:**
 - These Cooper pairs experience a collective quantum state where they move through the material without scattering, leading to zero electrical resistance. **This property allows for perfect conductivity.**
3. **Energy Gap and Condensation:**
 - BCS theory explains the existence of an energy gap, representing the minimum energy required to break the Cooper pairs. Below this gap, a large number of Cooper pairs condense into the lowest energy state, leading to superconductivity.
4. **Coherence and Long-Range Order:**
 - The theory describes how these paired electrons exhibit coherence over long distances, maintaining phase correlation throughout the material, which is essential for superconductivity.
5. **Critical Temperature (T_c):**
 - *The BCS theory predicts a critical temperature (T_c) below which a material becomes superconducting. This temperature depends on the strength of electron-phonon interactions.*
6. **Validity and Applications:**
 - BCS theory has successfully explained various experimental observations in superconductors. It's a cornerstone in understanding the properties and behaviour of superconducting materials.

The BCS theory significantly advanced the understanding of superconductivity, providing a theoretical framework for explaining the phenomenon and guiding the discovery and development of superconducting materials. Its concepts continue to influence research in condensed matter physics and material science, shaping our understanding of quantum phenomena in materials.

Josephson Junction:

When two superconductors are separated by thin insulating layer, an intriguing phenomenon occurs: Cooper pairs of electrons can tunnel through this junction (thin insulating layer). **This junction is called Josephson junction, a thin insulating layer between the two conductors.**

The wave function of the Cooper pairs on either side of the junction penetrates the insulating layer with exponentially decreasing amplitudes.

DC Josephson effect: In DC Josephson effect an electric current, tied to the phase difference between these wave functions, can flow through the junction even in the absence of voltage and is given by.

$$I_J = I_{max} \sin \phi \dots \dots \dots (1)$$

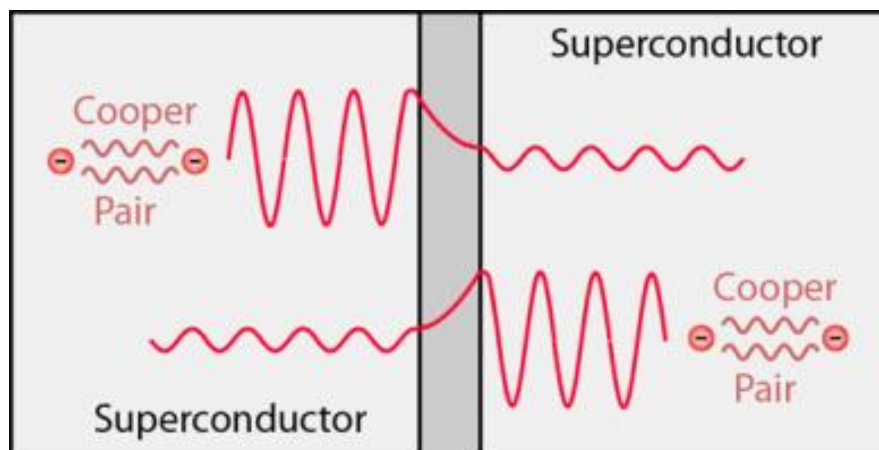
Where ϕ , is the phase difference between the two wave functions of the cooper pairs on either side of the junction, I_{max} is the maximum junction current depends on the thickness of the insulating layer.

AC Josephson effect: When voltage is applied across the junction the phase difference ϕ ,

increases with time at the rate of $\nu = \frac{d\phi}{dt} = \frac{2eV}{h}$, and as a result I_J

varies sinusoidally with time. This phenomenon is called AC Josephson effect. Where ν

is called the oscillation frequency.



Applications of Superconductors:

Superconductors have a broad range of applications across various fields, owing to their unique properties of zero electrical resistance and the ability to expel magnetic fields. Some prominent applications include:

Medical Applications:

1. Magnetic Resonance Imaging (MRI):

- Superconducting magnets are integral components in MRI machines, producing strong and stable magnetic fields necessary for detailed imaging of soft tissues and organs.

Power Generation and Transmission:

2. Electricity Transmission:

- Superconducting materials can carry electricity without any loss due to resistance, potentially revolutionizing power transmission, reducing energy wastage in transmission lines.

Scientific Research:

3. Particle Accelerators:

- Superconducting magnets are used in particle accelerators like the Large Hadron Collider (LHC) to guide and steer particle beams at high energies.

Transportation:

4. Magnetic Levitation (Maglev) Trains:

- Superconducting magnets enable Maglev trains to levitate above the tracks, reducing friction and allowing for high-speed, energy-efficient transportation.

Energy Storage:

5. Magnetic Energy Storage:

- Superconducting magnetic energy storage systems can store large amounts of energy with minimal losses, offering solutions for grid-scale energy storage.

Electronics and Computing:

6. Quantum Computing:

- Superconducting qubits are used in some quantum computers due to their ability to maintain quantum states for longer durations, aiding in more stable and scalable quantum systems.

Scientific Instrumentation:

7. Particle Detectors:

- Superconducting detectors are used in high-energy physics experiments and cosmic ray research due to their sensitivity to particles.

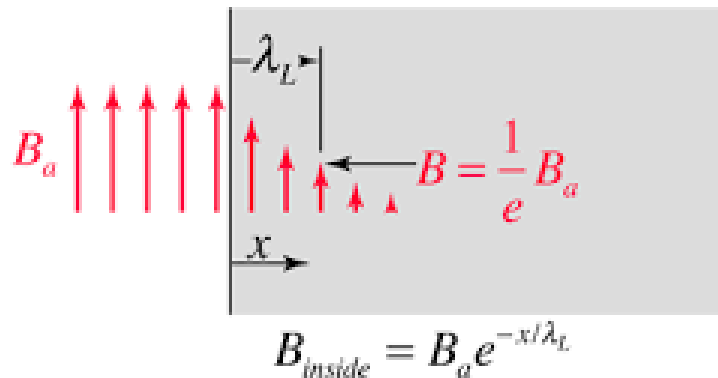
Sensors and Detectors:

8. Magnetic Sensors:

- Superconducting materials are utilized in highly sensitive magnetic sensors for various applications, including geophysical exploration and non-destructive testing.

The applications of superconductors continue to evolve, with ongoing research focused on discovering new superconducting materials, improving their properties, and finding ways to operate them at higher temperatures, which would expand their utility in numerous fields.

London Penetration Depth:



The London penetration depth is the distance required to fall to $1/e$ times the externally applied field B_a .

Supposed a superconductor is placed in an external magnetic field, an induced super current (like eddy current in normal metal conductor) starts flowing in such a direction, that the generated magnetic field due to this current opposes the external magnetic field B_a . If the applied magnetic field B_a is weak the magnetic flux is totally expelled out from the superconductor.

But in reality it is observed that inside the superconductor the magnetic field does not abruptly zero but it decays exponentially and it penetrates a finite distance from the surface of the superconductor to inside it as shown in the above figure. **The distance required to fall to $1/e$ times the value of external magnetic field B_a , is called the London penetration depth λ_L .**

Since the developed super current in super conductor is due to the electron's motion having charge $-e$ and mass m . Further since the electron's motion in superconductor is possible only if there will be an induced electric field E . Therefore from the Newton's second law of motion

$$F = m \frac{dv_s}{dt} = -eE \dots \dots \dots (1)$$

Where $\mathbf{v}_s$ is the velocity of the superconducting current carriers,

Therefore the current density can be written as

$$\mathbf{J}_s = -n_s \mathbf{v}_s e \dots \dots \dots (2)$$

Where n_s superconducting charge density.

Differentiating equation (2) with respect to t

$$\frac{d\mathbf{J}_s}{dt} = -e n_s \frac{d\mathbf{v}_s}{dt} \dots \dots \dots (3)$$

From equation (1) and (2), we have

$$\frac{d\mathbf{J}_s}{dt} = \frac{e^2 n_s E}{m} \dots \dots \dots (4)$$

Equation (4) is known as first London equation.

Taking curl of equation (4) we have

$$\vec{\nabla} \times \frac{d\vec{J}_s}{dt} = \frac{e^2 n_s}{m} \vec{\nabla} \times \vec{E} \dots \dots \dots (5)$$

From Maxwell's equation we know that

$$\vec{\nabla} \times \vec{E} = -\frac{d\vec{B}}{dt} \dots \dots \dots (6)$$

From equations (5) and (6) we can write

$$\begin{aligned} \frac{d}{dt} (\vec{\nabla} \times \vec{J}_s) &= -\frac{e^2 n_s}{m} \frac{d\vec{B}}{dt} \\ \Rightarrow \frac{d}{dt} \left\{ (\vec{\nabla} \times \vec{J}_s) + \frac{e^2 n_s}{m} \vec{B} \right\} &= \mathbf{0} \dots \dots \dots (7) \end{aligned}$$

In order to obtain Meissner effect ($B=0$ inside the superconductor), London has removed the time derivative and postulated the new equations.

$$(\vec{\nabla} \times \vec{J}_s) + \frac{e^2 n_s}{m} \vec{B} = \mathbf{0} \dots \dots \dots (8)$$

Since from the Maxwell's equation we know that

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}_s \dots \dots \dots (9)$$

From equation (8) and (9)

$$(\vec{\nabla} \times (\vec{\nabla} \times \vec{B})) + \frac{\mu_0 e^2 n_s}{m} \vec{B} = \mathbf{0}$$

$$\vec{\nabla}(\vec{\nabla} \cdot \vec{B}) - \nabla^2 \vec{B} + \frac{\vec{B}}{\lambda_L^2} = \mathbf{0}$$

$$\nabla^2 \vec{B} - \frac{\vec{B}}{\lambda_L^2} = \mathbf{0} \dots \dots \dots (10)$$

In one dimension we can write the above equation as

$$\frac{d^2 B}{dx^2} - \frac{B}{\lambda_L^2} = 0$$

Solution of the above equation

$$B = B_a e^{-\frac{x}{\lambda_L}}$$

Where $\lambda_L = \sqrt{\frac{m}{\mu_0 e^2 n_s}}$ is called the London Penetration depth.

μ_0 = permeability of free space